

going to move toward the brain rather than elsewhere; and there is no need to imagine any reason for this other than that the arteries that carry these parts of the blood there are those that come from the heart in the straightest line of all, and that, according to the laws of mechanics (which are the same as those of nature), when a number of things tend to move together in the same direction, where there is not enough room for all of them, as when the parts of the blood leaving the left cavity of the heart tend toward the brain, the weakest and least agitated must be pushed aside by the strongest which by this means arrive there alone. 55

I had provided a sufficiently detailed explanation for all these things in the treatise that I had previously intended to publish. And then I had shown what the constitution of the nerves and muscles of the human body must be in order to make the animal spirits within them have the force to move its members, as when one observes that heads, shortly after being severed, still move about and bite the earth, even though they are no longer alive. I had also shown what changes must take place in the brain in order to cause wakefulness, sleep, and dreams; how light, sounds, odors, tastes, heat, and all the other qualities of external objects can imprint various ideas there through the mediation of the senses; how hunger, thirst, and the other internal passions can also send their ideas there; what part of them needs to be taken there for the common sense, where these ideas are received, for the memory, which preserves them, and for the imagination, which can change them in various ways and compose new ones out of them, and, by the same means, distributing the animal spirits into the muscles, make the members of this body move in as many different ways (and in a manner appropriate to the objects that present themselves to the senses and to the internal passions that are in the body), as our own bodies can, without their being guided by the will. This will in no way seem strange to those who are cognizant of how many different automata or moving machines the ingenuity of men can make, without, in doing so, using more than a very small number of parts, in comparison with the great multitude of bones, muscles, nerves, arteries, veins, and all the other parts which are in the body of each animal. For they will regard this body as a machine which, having been made by the hands of God, is incomparably better ordered and has within itself movements far more wondrous than any of those that can be invented by men. 56

And I paused here in particular in order to show that, if there were such machines having the organs and the shape of a monkey or of some other animal that lacked reason, we would have no way of recognizing that they were not entirely of the same nature as these animals; whereas, if there were any such machines that bore a resemblance to our bodies

and imitated our actions as far as this is practically feasible, we would always have two very certain means of recognizing that they were not at all, for that reason, true men. The first is that they could never use words or other signs, or put them together as we do in order to declare our thoughts to others. For one can well conceive of a machine being so made that it utters words, and even that it utters words appropriate to the bodily actions that will cause some change in its organs (such as, if one touches it in a certain place, it asks what one wants to say to it, or, if in another place, it cries out that one is hurting it, and the like). But it could not
57 arrange its words differently so as to respond to the sense of all that will be said in its presence, as even the dumbest men can do. The second means is that, although they might perform many tasks very well or perhaps better than any of us, such machines would inevitably fail in other tasks; by this means one would discover that they were acting not through knowledge but only through the disposition of their organs. For while reason is a universal instrument that can be of help in all sorts of circumstances, these organs require some particular disposition for each particular action; consequently, it is for all practical purposes impossible for there to be enough different organs in a machine to make it act in all the contingencies of life in the same way as our reason makes us act.

Now by these two means one can also know the difference between men and beasts. For it is rather remarkable that there are no men so dull and so stupid (excluding not even the insane), that they are incapable of arranging various words together and of composing from them a discourse by means of which they might make their thoughts understood, and that, on the other hand, there is no other animal at all, however perfect and pedigreed it may be, that does the like. This does not happen because they lack the organs, for one sees that magpies and parrots can utter words just as we can, and yet they cannot speak as we do, that is to say, by testifying to the fact that they are thinking about what they are saying; on the other hand, men born deaf and dumb, who are deprived of the
58 organs that aid others in speaking just as much as, or more than beasts, are wont to invent for themselves various signs by means of which they make themselves understood to those who, being with them on a regular basis, have the time to learn their language. And this attests not merely to the fact that beasts have less reason than men but that they have none at all. For it is obvious it does not need much to know how to speak; and since we notice as much inequality among animals of the same species as among men, and that some are easier to train than others, it is unbelievable that a monkey or a parrot that is the most perfect of its species would not equal in this respect one of the most stupid children or at least a child with a disordered brain, if their soul were not of a nature entirely different

from our own. And we should not confuse words with the natural movements that attest to the passions and can be imitated by machines as well as by animals. Nor should we think, as did some of the ancients, that beasts speak, although we do not understand their language, for if that were true, since they have many organs corresponding to our own, they could make themselves as well understood by us as they are by their fellow creatures. It is also a very remarkable phenomenon that, although there are many animals that show more skill than we do in some of their actions, we nevertheless see that they show none at all in many other actions. Consequently, the fact that they do something better than we do does not prove that they have any intelligence, for, were that the case, they would have more of it than any of us and would excel us in everything. 59 But rather it proves that they have no intelligence at all, and that it is nature that acts in them, according to the disposition of their organs—just as we see that a clock composed exclusively of wheels and springs can count the hours and measure time more accurately than we can with all our carefulness.

After that, I described the rational soul and showed that it can in no way be derived from the potentiality of matter, as can the other things I have spoken of, but rather that it must be expressly created; and how it is not enough for it to be lodged in the human body like a pilot in his ship, unless perhaps in order to move its members, but rather that it must be more closely joined and united to the body in order to have, in addition to this, feelings and appetites similar to our own, and thus to constitute a true man. As to the rest, I elaborated here a little on the subject of the soul because it is of the greatest importance; for, after the error of those who deny the existence of God (which I think I have sufficiently refuted), there is none at all that puts weak minds at a greater distance from the straight path of virtue than to imagine that the soul of beasts is of the same nature as ours, and that, as a consequence, we have nothing to fear or to hope for after this life any more than do flies and ants. On the other hand, when one knows how different they are, one understands much better the arguments which prove that our soul is of a nature entirely independent of the body, and consequently that it is not subject to die with it. Then, since we do not see any other causes at all for its destruction, 60 we are naturally led to judge from this that it is immortal.

PART SIX

But it is now three years since I arrived at the end of the treatise that contains all these things and began to review it in order to put it into the hands of a printer, when I learned that some people to whom I defer and